

KCPC Workshop 2

Solutions

1 Psychological Strategies

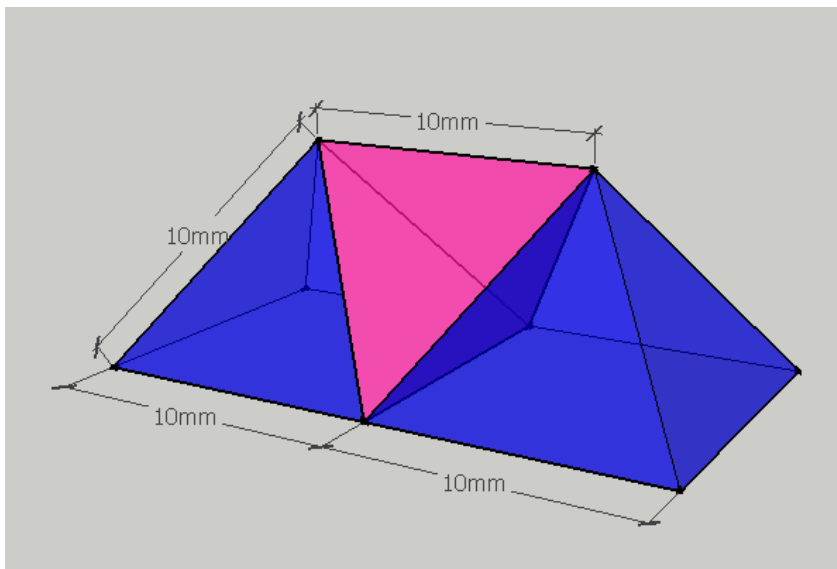
1.1 Pat's Dilemma

Solution: This is unsolvable if we limit ourselves to two-dimensional space. Once liberated from Flatland, we get a nice solution: The sword fits into a $1 \times 1 \times 1$ -meter box, with a long diagonal of $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} > 1.69$ meters.

1.2 Thinking in 3D

Answer: 5 faces.

Explanation: The square pyramid has 5 faces (1 square + 4 equilateral triangles) and the regular tetrahedron has 4 triangular faces. Gluing them along one triangular face removes that face from each solid, so $5 + 4 - 2 = 7$ at first. But because all edges are length 1 the pyramid's lateral faces are equilateral, and the dihedral angle between two adjacent lateral faces equals the dihedral angle of a regular tetrahedron (both are $\arccos(1/3)$). So the two faces of the tetrahedron adjacent to the glued triangle lie in the same planes as the two lateral triangles adjacent to the glued triangle; each such pair merges into one planar quadrilateral. That reduces the count by 2 more, giving $7 - 2 = 5$ faces in the final solid.



2 Wishful Thinking and Make it Easier

2.1 Long Sequence

Answer: $a_{6767} = 6767^2$.

This equation is hard to understand without experimentation. Let's try to build up some values of a_n . First, we *keep things simple* and try $m = n = 0$, which yields $a_0 + a_0 = a_0$, so $a_0 = 0$. Now use the known indices 0, 1: plug $m = 1, n = 0$ into the equation and we have $2a_1 = \frac{1}{2}(a_2 + a_0) = a_2/2$, so $a_2 = 4a_1 = 4$. More generally, if we just fix $n = 0$, we have $2a_m = a_{2m}/2$, or $a_{2m} = 4a_m$. Now let's plug in $m = 2, n = 1$:

$$a_{2+1} + a_{2-1} = \frac{1}{2}(a_4 + a_2).$$

Since $a_4 = 4a_2 = 4 \times 4 = 16$, we have $a_3 + a_1 = \frac{1}{2}(16 + 4) = 10$, so $a_3 = 9$. At this point, we are ready to venture the conjecture that $a_n = n^2$. If this conjecture is true, a likely method of proof will be *mathematical induction*. See the next section for details on induction.

2.2 An Algorithm?

Solution: $f(n)$ equals the number of 1's in the binary representation of n (the popcount of n).

3 Induction

Proof: The base case, $n_0 = 4$, is obviously true: $4! > 2^4$. Now assume that $n! > 2^n$ for some n . We wish to use this to prove the "next" case; i.e., we wish to prove that $(n+1)! > 2^{n+1}$. Let's think strategically: the left-hand side of the inductive hypothesis is $n!$, and the left-hand side of the "goal" is $(n+1)!$. How to get from one to the other? Multiply both sides of the inductive hypothesis by $n+1$, of course. Multiplying both sides of an inequality by a positive number doesn't change its truth, so we get

$$(n+1)! > 2^n(n+1).$$

This is almost what we want, for the right-hand side of the "goal" is $2^{n+1} = 2^n \times 2$, and $n+1$ is certainly greater than 2; i.e.,

$$(n+1)! > 2^n(n+1) > 2^n \times 2 = 2^{n+1}.$$

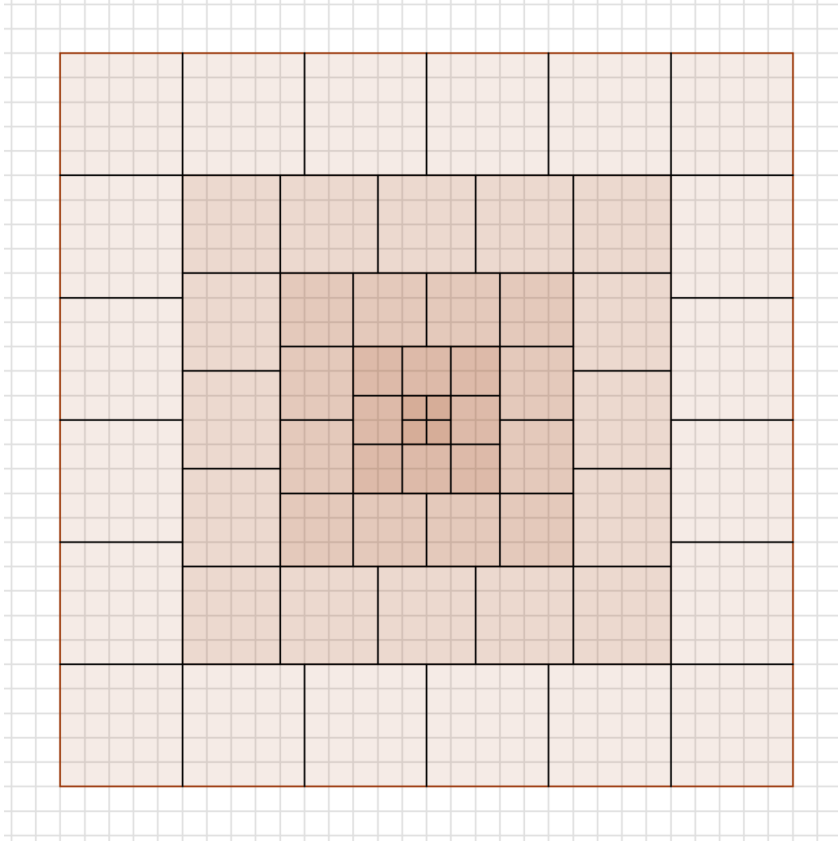
Induction arguments can be rather subtle. Sometimes it is not obvious what plays the role of the "index" n . Sometimes the crux move is a clever choice of this variable, and/or a neat method of traveling between $P(n)$ and $P(n+1)$.

4 Draw a Picture

We are trying to prove that

$$\sum_{i=1}^n i^3 = \left(\sum_{i=1}^n i \right)^2 = \frac{n^2(n+1)^2}{4}.$$

Solution: A visual illustration of the proof for $n = 5$ is shown below.



The number of squares of side n is seen to be $4n$. To go from n to $n+1$, a ring of $4(n+1)$ squares of side $n+1$ is to be added: 1) 4 for the one at each corner, $4n$ for the ones that abut the sides of the $n+1$ squares of side n .

The length of one side is given by: $S = 2(1 + 2 + 3 + \cdots + n)$. The length of one side is also given by: $S = n(n+1)$.

The area is therefore given in two ways as:

$$A = 4(1 + 2 + \cdots + n)^2 = (n(n+1))^2,$$

$$A = 4 \times 1 \times 1^2 + 4 \times 2 \times 2^2 + 4 \times 3 \times 3^2 + \cdots + 4 \times n \times n^2 = 4 \sum_{i=1}^n i^3.$$

The result follows by equating these two expressions for the area.

More interesting proofs for this problem: [here](#).

5 Recast

Solution: At first, this seems like a geometric/combinatorial problem with many cases and subcases. But it is really just a question about counting colors. The two corners that

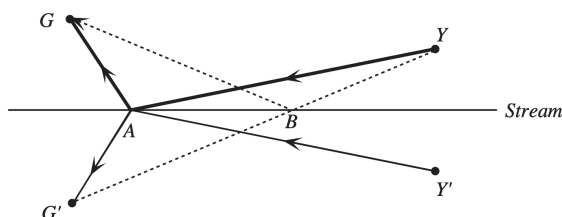
were removed were both (without loss of generality) white, so the shape we are interested in contains 32 black and 30 white squares. Yet any domino, once it is placed, will occupy exactly one black and one white square. The 31 dominos thus require 31 black and 31 white squares, so tiling is impossible.

The idea of introducing coloring to reformulate a problem is quite old. Nevertheless, it took several years before anyone thought to use this method on the Gallery Problem. This problem was first proposed by Victor Klee in 1973, and solved shortly thereafter by Václav Chvátal. However, his proof was rather complicated. In 1978, S. Fisk discovered the elegant coloring argument. If you want to read more about this beautiful solution, see [here](#).

6 Symmetry

6.1 Grandma's Cabin

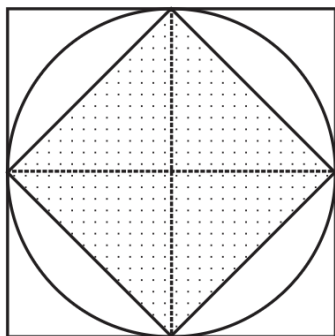
Solution: First, draw a picture! Label your location by Y and Grandma's by G . Certainly, this problem can be done with calculus, but it is very ugly (you need to differentiate the sum of two radicals, for starters). The problem appears to have no symmetry in it, but the stream is practically begging you to reflect across it! Draw a sample path (shown below as YA followed by AG) and look at its reflection. We call the reflections of your house and Grandma's house Y' and G' , respectively.



While you are carrying water to Grandma, your duplicate in an alternate universe is doing the same, only south of the stream. Notice that $AG = AG'$, so the length of your path would be unchanged if you visited the reflected Grandma instead of the real one. Since the shortest distance between two points is the straight line YBG' , the optimal path will be YB followed by BG . Its length is the same as the length of YBG' , which is just the hypotenuse of a right triangle with legs 5 and 12 miles. Hence our answer is 13 miles.

6.2 Geometric

The problem can certainly be solved algebraically (let x equal the length of the small square, then use the Pythagorean theorem, etc.), but there is a nicer approach. The original diagram is full of symmetries. We are free to rotate and/or reflect many shapes and still preserve the areas of the two squares. How do we choose from all these possibilities? We need to use the hypothesis that the objects are inscribed in one another. If we rotate the small square by 45 degrees, its vertices now line up with the points of tangency between the circle and the large square, and instantly the solution emerges.



The small square obviously has half the area of the larger!

6.3 In Probability

Solution for the first: Think of the 4 aces as cutting the ordered deck into 5 segments: the cards before the first ace, the three gaps between consecutive aces, and the cards after the last ace. By symmetry (the aces are equally likely to fall anywhere), all five segments have the same expected length. If we include a fictitious marker at the very end of the deck, the total number of "items" split among those five segments is 53 (the 52 real cards plus the marker). So the expected size of each segment is $53/5 = 10.6$. In particular, the expected number of cards before (and including) the first ace is 10.6, so on average you draw 10.6 cards to see the first ace.

Solution for the second: 3B1B video explanation.

7 The Extreme Principle

Solution: After experimenting, it seems that if the points do not all lie on a single line, there cannot be finitely many of them. It may be possible to prove this with many complicated diagrams showing that "you can always draw a new point," but this isn't easy. The extreme principle comes to the rescue: Assume that the points do not all lie on a line. Then they form at least one triangle. Consider the triangle of smallest area. Two of its vertices are the same color, so between them is a point of the other color, but this forms a smaller triangle—a contradiction!

Note how the extreme principle immediately cracked the problem. It was so easy that it almost seemed like cheating. The structure of the argument used is a typical one: Work by contradiction; assume that whatever you wish to prove is not true. Then look at the minimal (or maximal) element and develop an argument that creates a smaller (or larger) element, which is the desired contradiction. As long as a set of real numbers is finite, it will have a minimum and a maximum element. For infinite sets, there may be no extreme values (for example, consider the infinite set $1, 2, 1/2, 2^2, 1/2^2, 2^3, 1/2^3, \dots$, which has neither a minimum nor a maximum element), but if the set consists of positive integers, we can use the Well-Ordering Principle:

Every non-empty set of positive integers has a least element.